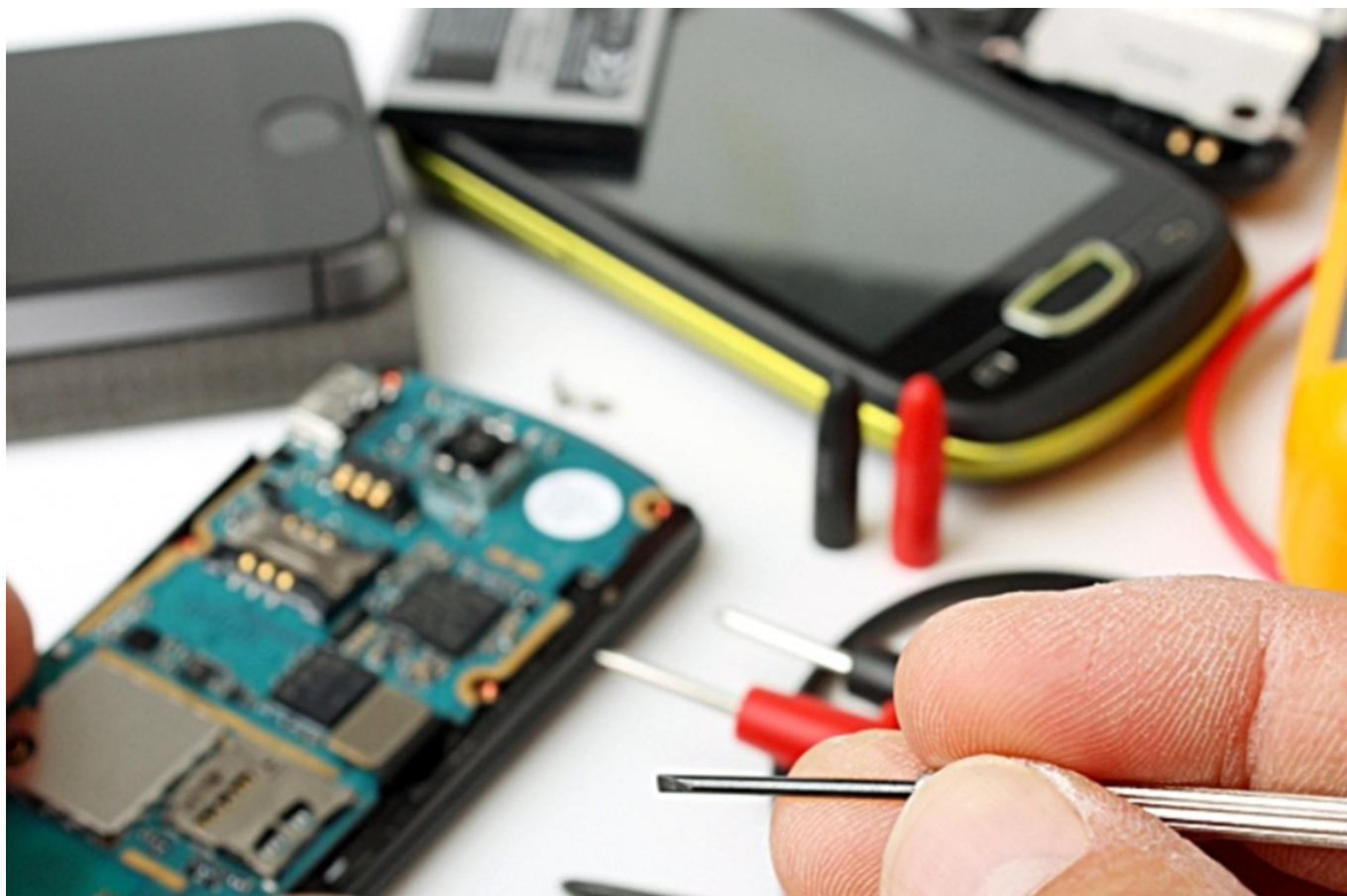


Qualification Pack



Repair and Maintenance Assistant "Smart Phones"

QP Code: NIE/ELE/Q0803

Version: 1.0

NSQF Level: 3

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Qualification Pack

NIE/ELE/Q0803: Repair and Maintenance Assistant "Smart Phones"

Brief Job Description

The Smartphone Repair and Troubleshooting program equips participants with comprehensive knowledge and skills essential for diagnosing, repairing, and troubleshooting hardware and software issues in smartphones. Participants will delve into the fundamentals of basic electronics and mobile phone technology, gaining an in-depth understanding of various hardware and software components crucial for smartphone functionality. The program focuses on practical techniques for hardware repair, including component replacement, soldering, and reworking, as well as troubleshooting software issues ranging from operating system errors to app-related problems.

Personal Attributes

Students will develop meticulous attention to detail, problem-solving abilities, and technical proficiency in both hardware and software aspects. They will learn to exhibit patience and precision while diagnosing and repairing complex issues, and uphold professionalism through ethical practices and effective communication. The course also encourages a proactive approach to continuous learning and adaptability, essential for keeping pace with evolving technologies and industry standards.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ELE/N0810: Foundations of Mobile Phones Technology](#)
2. [NIE/ELE/N0811: Smartphone Hardware Repair](#)
3. [NIE/ELE/N0812: Software Repair](#)
4. [NIE/ELE/N0813: Advanced Repair and Business Skills](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
6. [NIE/ELE/N0815: Implementation of Repair & Maintenance Project](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Electronics Assembly and Services
Occupation	Electronics Equipment Services

Qualification Pack

Country	India
NSQF Level	3
Credits	11
Aligned to NCO/ISCO/ISIC Code	7422.2301
Minimum Educational Qualification & Experience	10th grade pass with NA of experience OR 8th grade pass with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level with 1.5 years of experience
Minimum Level of Education for Training in School	8th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	30/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-03-EH-02595-2024-V1-NIELIT
NQR Version	1

Qualification Pack

NIE/ELE/N0810: Foundations of Mobile Phones Technology

Description

Foundations of Mobile Phone Technology provides students with essential knowledge and skills in mobile phone technology and repair. Students will learn fundamental concepts such as electricity, circuits, voltage, current, and resistance, and understand key electronic components like capacitors, resistors, and transistors. They will explore the historical development of mobile phones, distinguishing between generations from 2G to 5G, and comparing feature phones to smartphones. The module also covers the Android and iOS operating systems, internal mobile phone components, and essential repair tools. Emphasis is placed on Electrostatic Discharge (ESD) protection and safe handling practices to prevent damage during repairs. This module equips students to analyze, troubleshoot, and repair mobile phones effectively and safely.

Scope

The scope covers the following :

- This module provides students with a thorough understanding of mobile phone technology and repair, covering fundamental concepts such as electricity, circuits, voltage, current, and resistance. It explores the roles of components like capacitors, resistors, and transistors, and traces the historical development of mobile technology from 2G to 5G. Students will learn to differentiate between feature phones and smartphones and compare Android and iOS operating systems. They will also identify and describe internal mobile phone components, including batteries, displays, motherboards, cameras, and sensors. Practical skills include using basic repair tools, understanding Electrostatic Discharge (ESD) protection, and applying safe handling practices to ensure an effective work environment.

Elements and Performance Criteria

Basic Electronics

To be competent, the user/individual on the job must be able to:

- PC1.** Define and explain the concepts of electricity, circuits, voltage, current, and resistance.
- PC2.** Identify and describe the functions of capacitors, resistors, and transistors in electronic circuits.

Mobile Phone Technology

To be competent, the user/individual on the job must be able to:

- PC3.** Trace the historical development of mobile phones and Overview of Generations of Mobile Technology (2G, 3G, 4G, 5G)
- PC4.** Types of Mobile Phones: Feature Phones, Smartphones and describe the Operating Systems: Android and iOS

Phone Anatomy

To be competent, the user/individual on the job must be able to:

- PC5.** Identify and describe the internal components of a mobile phone, including the battery, display, motherboard, camera, and sensors.
- PC6.** Recognize basic tools used in mobile phone repair

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Safety and Precautions

To be competent, the user/individual on the job must be able to:

- PC7.** Importance of Electrostatic Discharge Protection and Safe Handling of Tools and Equipment
- PC8.** Establish and maintain a safe work environment conducive to effective and efficient repair processes.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Grasp the core principles of electricity, circuits, and key electronic components (capacitors, resistors, transistors), and understand their roles in mobile phone technology.
- KU2.** Comprehend the evolution of mobile phones from their inception to the present, including the development of different generations of mobile technology (2G, 3G, 4G, 5G) and the distinct characteristics of feature phones versus smartphones.
- KU3.** Identify and describe the internal components of mobile phones, such as the battery, display, motherboard, camera, and sensors, and understand their specific functions and contributions to the overall operation of the device & Recognize basic tools used in mobile phone repair.
- KU4.** Appreciate the importance of Electrostatic Discharge (ESD) protection to prevent damage to electronic components and apply effective measures to mitigate ESD risks.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop a solid understanding of fundamental electronics, including concepts like electricity, circuits, voltage, current, and resistance, along with knowledge of key components such as capacitors, resistors, and transistors.
- GS2.** Gain insight into the history and evolution of mobile phone technology, different generations (2G, 3G, 4G, 5G), and the comparative features of feature phones versus smartphones, as well as operating systems like Android and iOS.
- GS3.** Acquire the skills to identify and understand the functions of internal mobile phone components, including batteries, displays, motherboards, cameras, and sensors and learn to use basic tools effectively for mobile phone repair,
- GS4.** Understand the importance of Electrostatic Discharge (ESD) protection and implement appropriate measures to safeguard electronic components during repair.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basic Electronics</i>	50	23	-	-
PC1. Define and explain the concepts of electricity, circuits, voltage, current, and resistance.	-	-	-	-
PC2. Identify and describe the functions of capacitors, resistors, and transistors in electronic circuits.	-	-	-	-
<i>Mobile Phone Technology</i>	-	-	-	-
PC3. Trace the historical development of mobile phones and Overview of Generations of Mobile Technology (2G, 3G, 4G, 5G)	-	-	-	-
PC4. Types of Mobile Phones: Feature Phones, Smartphones and describe the Operating Systems: Android and iOS	-	-	-	-
<i>Phone Anatomy</i>	-	-	-	-
PC5. Identify and describe the internal components of a mobile phone, including the battery, display, motherboard, camera, and sensors.	-	-	-	-
PC6. Recognize basic tools used in mobile phone repair	-	-	-	-
<i>Safety and Precautions</i>	-	-	-	-
PC7. Importance of Electrostatic Discharge Protection and Safe Handling of Tools and Equipment	-	-	-	-
PC8. Establish and maintain a safe work environment conducive to effective and efficient repair processes.	-	-	-	-
NOS Total	50	23	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0810
NOS Name	Foundations of Mobile Phones Technology
Sector	Electronics
Sub-Sector	
Occupation	Electronics Equipment Services
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0811: Smartphone Hardware Repair

Description

Smartphone Hardware Repair equips students with essential skills for diagnosing and repairing common smartphone issues. It covers troubleshooting techniques for problems such as charging issues, display faults, water damage, and software glitches, utilizing diagnostic tools and visual inspection methods. The module emphasizes safe disassembly and assembly practices, including handling various phone models and their specific disassembly processes. Students will gain hands-on experience in replacing key components like displays, batteries, charging ports, microphones, and speakers, and will learn soldering and desoldering techniques for component replacement. Additionally, the module delves into basic circuit analysis and troubleshooting, teaching students to understand phone circuits and identify faulty components using multimeters and other diagnostic tools.

Scope

The scope covers the following :

- Smartphone Hardware Repair provides a comprehensive understanding of the skills necessary for effective smartphone maintenance and repair. It covers troubleshooting techniques to identify common issues such as charging problems, display faults, water damage, and software glitches, employing both diagnostic tools and visual inspection methods. The module emphasizes safe disassembly and reassembly practices for various phone models, ensuring students can handle the intricacies of different devices. Practical skills in replacing key components like displays, batteries, charging ports, microphones, and speakers are taught alongside soldering and desoldering techniques. Additionally, the module focuses on basic circuit analysis and troubleshooting, enabling students to understand phone circuitry and accurately identify faulty components using multimeters and other diagnostic tools.

Elements and Performance Criteria

Troubleshooting Techniques

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and diagnose common smartphone problems, including charging issues, display problems, water damage, and software glitches.
- PC2.** Use diagnostic tools and conduct visual inspections to pinpoint faults and issues in smartphones.

Disassembly and Assembly

To be competent, the user/individual on the job must be able to:

- PC3.** Demonstrate safe disassembly and assembly techniques for smartphones to avoid damage to components.
- PC4.** Understand and apply disassembly and reassembly processes for various phone models, recognizing the unique aspects of each model.

Component Replacement

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC5.** Replace common smartphone components such as displays, batteries, charging ports, microphones, and speakers with precision and care.
- PC6.** Utilize soldering and desoldering techniques effectively for component replacement, ensuring secure and functional connections.

Circuit Analysis and Troubleshooting

To be competent, the user/individual on the job must be able to:

- PC7.** Understand the basic circuitry of smartphones, including the function and layout of key components.
- PC8.** Identify and diagnose faulty components within smartphone circuits using multimeters and other diagnostic tools to ensure accurate repairs.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand how to identify and diagnose common smartphone problems, such as charging issues, display problems, water damage, and software glitches.
- KU2.** Learn to effectively use diagnostic tools and conduct visual inspections to accurately pinpoint and address these issues.
- KU3.** Gain proficiency in safely disassembling and assembling smartphones, including an understanding of the processes for various phone models.
- KU4.** Develop practical skills for replacing common smartphone components like displays, batteries, charging ports, microphones, and speakers, along with mastering soldering and desoldering techniques.
- KU5.** Acquire a basic understanding of smartphone circuitry, including the layout and function of key components and Learn to identify faulty components using multimeters and other diagnostic tools to ensure precise and effective repairs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop critical thinking and analytical skills to identify and diagnose common smartphone issues such as charging problems, display faults, water damage, and software glitches.
- GS2.** Gain proficiency in using diagnostic tools and conducting visual inspections to accurately determine the cause of issues and formulate effective repair strategies.
- GS3.** Develop expertise in replacing critical components such as displays, batteries, charging ports, microphones, and speakers, and refine soldering and desoldering techniques to ensure high-quality repairs.
- GS4.** Enhance your ability to understand and analyze smartphone circuits, recognizing the layout and function of key components.
- GS5.** Learn to use multimeters and other diagnostic tools to identify and troubleshoot faulty components, ensuring accurate and reliable repairs in a professional setting.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Troubleshooting Techniques</i>	80	35	-	-
PC1. Identify and diagnose common smartphone problems, including charging issues, display problems, water damage, and software glitches.	-	-	-	-
PC2. Use diagnostic tools and conduct visual inspections to pinpoint faults and issues in smartphones.	-	-	-	-
<i>Disassembly and Assembly</i>	-	-	-	-
PC3. Demonstrate safe disassembly and assembly techniques for smartphones to avoid damage to components.	-	-	-	-
PC4. Understand and apply disassembly and reassembly processes for various phone models, recognizing the unique aspects of each model.	-	-	-	-
<i>Component Replacement</i>	-	-	-	-
PC5. Replace common smartphone components such as displays, batteries, charging ports, microphones, and speakers with precision and care.	-	-	-	-
PC6. Utilize soldering and desoldering techniques effectively for component replacement, ensuring secure and functional connections.	-	-	-	-
<i>Circuit Analysis and Troubleshooting</i>	-	-	-	-
PC7. Understand the basic circuitry of smartphones, including the function and layout of key components.	-	-	-	-
PC8. Identify and diagnose faulty components within smartphone circuits using multimeters and other diagnostic tools to ensure accurate repairs.	-	-	-	-
NOS Total	80	35	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0811
NOS Name	Smartphone Hardware Repair
Sector	Electronics
Sub-Sector	
Occupation	Electronics Equipment Services
NSQF Level	3
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0812: Software Repair

Description

This module provides a comprehensive understanding of smartphone software repair. Students will learn the fundamentals of Android and iOS operating systems, focusing on system settings, app management, and file systems. They will gain expertise in flashing ROMs and rooting Android phones, understanding the associated risks and benefits. The module also covers data recovery techniques, enabling students to recover data from damaged or formatted phones using specialized software. Additionally, students will learn to identify and remove malware and viruses, and implement secure practices for installing apps and updates, ensuring the overall security and optimal performance of smartphones.

Scope

The scope covers the following :

- This module focuses on the essential aspects of smartphone software repair. It covers the fundamentals of Android and iOS operating systems, including system settings, app management, and file systems. Students will explore techniques for flashing ROMs and rooting Android devices, along with understanding the associated risks and benefits. The module also addresses data recovery methods, teaching students how to retrieve data from damaged or formatted phones using specialized software. Additionally, it covers security practices, including identifying and removing malware and viruses, and ensuring secure installation of apps and updates to maintain optimal device performance and security.

Elements and Performance Criteria

Operating Systems

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate a thorough understanding of the basics of Android and iOS operating systems.
- PC2.** Effectively manage system settings, applications, and file systems on both Android and iOS devices.

Flashing and Rooting

To be competent, the user/individual on the job must be able to:

- PC3.** Apply techniques for flashing ROMs and rooting Android phones.
- PC4.** Evaluate and articulate the risks and benefits associated with flashing and rooting, ensuring informed decision-making.

Data Recovery

To be competent, the user/individual on the job must be able to:

- PC5.** Utilize effective techniques to recover data from damaged or formatted phones and Employ data recovery software proficiently to restore lost information.

Security and Malware

To be competent, the user/individual on the job must be able to:

- PC6.** Identify and remove malware and viruses from smartphones.

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PC7. Implement secure practices for installing apps and updates to safeguard device security and functionality.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the basic functionalities of Android and iOS operating systems, including their core features and user interfaces.
- KU2.** Gain proficiency in managing system settings, app installations, and file systems on both Android and iOS devices.
- KU3.** Learn the techniques for flashing ROMs and rooting Android phones, including the procedures involved and the impact on device functionality.
- KU4.** Comprehend the risks and benefits associated with these modifications, ensuring informed decisions and appropriate handling.
- KU5.** Develop expertise in recovering data from damaged or formatted phones by effectively using specialized data recovery software.
- KU6.** Understand the methods for identifying and removing malware and viruses, and implement secure practices for installing apps and updates to protect device security.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop the ability to manage and configure Android and iOS operating systems, including handling system settings, app installations, and file systems efficiently.
- GS2.** Acquire skills in data recovery from damaged or formatted phones, leveraging specialized software for effective restoration of lost information.
- GS3.** Cultivate expertise in identifying and removing malware and viruses, and applying secure practices for installing apps and updates to maintain device security.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operating Systems</i>	50	22	-	-
PC1. Demonstrate a thorough understanding of the basics of Android and iOS operating systems.	-	-	-	-
PC2. Effectively manage system settings, applications, and file systems on both Android and iOS devices.	-	-	-	-
<i>Flashing and Rooting</i>	-	-	-	-
PC3. Apply techniques for flashing ROMs and rooting Android phones.	-	-	-	-
PC4. Evaluate and articulate the risks and benefits associated with flashing and rooting, ensuring informed decision-making.	-	-	-	-
<i>Data Recovery</i>	-	-	-	-
PC5. Utilize effective techniques to recover data from damaged or formatted phones and Employ data recovery software proficiently to restore lost information.	-	-	-	-
<i>Security and Malware</i>	-	-	-	-
PC6. Identify and remove malware and viruses from smartphones.	-	-	-	-
PC7. Implement secure practices for installing apps and updates to safeguard device security and functionality.	-	-	-	-
NOS Total	50	22	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0812
NOS Name	Software Repair
Sector	Electronics
Sub-Sector	
Occupation	Electronics Equipment Services
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0813: Advanced Repair and Business Skills

Description

Advanced Repair and Business Skills is designed to provide comprehensive training in both advanced technical and business aspects of mobile phone repair. The Advanced Troubleshooting section focuses on handling complex phone issues and understanding schematics and component diagrams. The Business Management section covers essential topics such as establishing and managing a repair shop, implementing effective marketing and customer service strategies, and managing finances and inventory. The Professional Ethics section emphasizes the importance of honesty, data privacy, customer satisfaction, and maintaining integrity in business operations. Lastly, the Career Opportunities section explores job opportunities in the mobile phone repair industry, entrepreneurship, and relevant government initiatives and policies. This module aims to equip learners with the skills necessary to excel in the mobile phone repair industry and to successfully manage their own repair businesses.

Scope

The scope covers the following :

- Advanced Repair and Business Skills covers essential skills for the mobile phone repair industry. It includes advanced troubleshooting for complex phone issues and understanding schematics, as well as business management for setting up and running a repair shop, marketing, customer service, and managing finances and inventory. The module emphasizes professional ethics, focusing on honesty, data privacy, customer satisfaction, and business integrity. It also explores career opportunities, including job prospects, entrepreneurship, and relevant government initiatives. This module prepares individuals for both technical and business success in mobile phone repair.

Elements and Performance Criteria

Advanced Troubleshooting

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and diagnose intricate phone issues and apply appropriate troubleshooting techniques to resolve complex problems.
- PC2.** Interpret and read schematics and component diagrams accurately and utilize schematics to aid in troubleshooting and repair.

Business Management

To be competent, the user/individual on the job must be able to:

- PC3.** Plan and establish a mobile phone repair shop and Implement effective management practices for shop operations.
- PC4.** Develop and execute marketing strategies to attract customers.
- PC5.** Manage financial aspects, including budgeting, accounting, and financial planning and maintain and control inventory to ensure availability of necessary parts and supplies.

Professional Ethics

To be competent, the user/individual on the job must be able to:

- PC6.** Uphold principles of honesty and data privacy and ensure of customer satisfaction.
- PC7.** Maintain integrity in all business operations and interactions.

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Career Opportunities

To be competent, the user/individual on the job must be able to:

- PC8.** Explore various job opportunities within the mobile phone repair industry.
- PC9.** Develop a business plan for starting and managing a mobile phone repair business.
- PC10.** Understand and comply with relevant government initiatives and policies in the mobile phone repair field.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding intricate phone issues and applying effective troubleshooting techniques. Reading and interpreting schematics and component diagrams to diagnose and repair complex problems.
- KU2.** Knowledge of setting up and managing a mobile phone repair shop, including marketing strategies, customer service, financial management, and inventory control.
- KU3.** Comprehending the importance of honesty, data privacy, and customer satisfaction in business.
- KU4.** Exploring various job opportunities, entrepreneurship, and understanding government initiatives and policies relevant to the mobile phone repair industry.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Expertise in diagnosing and resolving intricate phone issues, and proficiency in reading and interpreting schematics and component diagrams.
- GS2.** Competence in setting up and managing a repair shop, including marketing, customer service, financial management, and inventory control.
- GS3.** Commitment to upholding honesty, data privacy, and customer satisfaction, and understanding career opportunities, entrepreneurship, and relevant government initiatives and policies.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Advanced Troubleshooting</i>	20	10	-	-
PC1. Identify and diagnose intricate phone issues and apply appropriate troubleshooting techniques to resolve complex problems.	-	-	-	-
PC2. Interpret and read schematics and component diagrams accurately and utilize schematics to aid in troubleshooting and repair.	-	-	-	-
<i>Business Management</i>	-	-	-	-
PC3. Plan and establish a mobile phone repair shop and Implement effective management practices for shop operations.	-	-	-	-
PC4. Develop and execute marketing strategies to attract customers.	-	-	-	-
PC5. Manage financial aspects, including budgeting, accounting, and financial planning and maintain and control inventory to ensure availability of necessary parts and supplies.	-	-	-	-
<i>Professional Ethics</i>	-	-	-	-
PC6. Uphold principles of honesty and data privacy and ensure of customer satisfaction.	-	-	-	-
PC7. Maintain integrity in all business operations and interactions.	-	-	-	-
<i>Career Opportunities</i>	-	-	-	-
PC8. Explore various job opportunities within the mobile phone repair industry.	-	-	-	-
PC9. Develop a business plan for starting and managing a mobile phone repair business.	-	-	-	-
PC10. Understand and comply with relevant government initiatives and policies in the mobile phone repair field.	-	-	-	-
NOS Total	20	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0813
NOS Name	Advanced Repair and Business Skills
Sector	Electronics
Sub-Sector	
Occupation	Electronics Equipment Services
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

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NIE/ELE/N0815: Implementation of Repair & Maintenance Project

Description

The Smartphone Repair and Troubleshooting program equips participants with comprehensive knowledge and skills essential for diagnosing, repairing, and troubleshooting hardware and software issues in smartphones. Participants will delve into the fundamentals of basic electronics and mobile phone technology, gaining an in-depth understanding of various hardware and software components crucial for smartphone functionality. The program focuses on practical techniques for hardware repair, including component replacement, soldering, and reworking, as well as troubleshooting software issues ranging from operating system errors to app-related problems.

Scope

The scope covers the following :

- The NOS (National Occupational Standards) Scope outlines the essential competencies required for a Repair and Maintenance Assistant in the smartphone industry. It defines the key tasks such as diagnosing faults, performing repairs, replacing components, and ensuring quality standards. Additionally, the scope includes adherence to workplace safety regulations, customer service skills, and familiarity with tools and software used in smartphone maintenance. This framework ensures a standardized skill set for professionals, enhancing efficiency and service quality in the industry.

Elements and Performance Criteria

Perform Repair and Maintenance Operations

To be competent, the user/individual on the job must be able to:

- PC1.** Dismantle and reassemble smartphones using appropriate tools.
- PC2.** Replace faulty components like screens, batteries, and motherboards.
- PC3.** Apply proper soldering and desoldering techniques when required.
- PC4.** Follow ESD (Electrostatic Discharge) safety measures during repair.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding smartphone hardware, software, operating systems, and common faults, along with effective troubleshooting and diagnostic techniques.
- KU2.** Knowledge of essential repair tools, safe handling practices, dismantling, reassembly, software updates, and quality control measures.
- KU3.** Familiarity with industry safety standards, reporting structures, customer service protocols, and proper documentation for repair services.
- KU4.** Awareness of warranty policies, intellectual property rights, e-waste management, business ethics, and efficient service management.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** Ability to read technical manuals, follow repair guidelines, and communicate effectively with customers. Clearly explain issues, provide repair estimates, and handle customer queries professionally.
- GS2.** Strong analytical and critical thinking skills to diagnose faults, determine appropriate repair solutions, and ensure efficient troubleshooting while maintaining quality and cost-effectiveness.
- GS3.** Ability to manage repair tasks, prioritize workload, maintain records, handle inventory, and follow workplace procedures to ensure smooth operations and customer satisfaction.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform Repair and Maintenance Operations</i>	-	-	-	30
PC1. Dismantle and reassemble smartphones using appropriate tools.	-	-	-	-
PC2. Replace faulty components like screens, batteries, and motherboards.	-	-	-	-
PC3. Apply proper soldering and desoldering techniques when required.	-	-	-	-
PC4. Follow ESD (Electrostatic Discharge) safety measures during repair.	-	-	-	-
NOS Total	-	-	-	30

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0815
NOS Name	Implementation of Repair & Maintenance Project
Sector	Electronics
Sub-Sector	
Occupation	Electronics Equipment Services
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria.

The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. competence acquired by the candidate can be obtained by conducting Theory, Practical assessment, Internal assessment. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number.
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are

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through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.
6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0810.Foundations of Mobile Phones Technology	50	23	0	0	73	20
NIE/ELE/N0811.Smartphone Hardware Repair	80	35	0	0	115	32

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0812.Software Repair	50	22	0	0	72	20
NIE/ELE/N0813.Advanced Repair and Business Skills	20	10	0	0	30	8
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	14
NIE/ELE/N0815.Implementation of Repair & Maintenance Project	-	-	-	30	30	6
Total	220	120	-	30	370	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

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Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf